# Bayesian Data Analysis of the German Tank Problem

## Theoretical Properties, Simulation Benchmarks, and Real-World Stress-Testing

Inferring Discrete Ceilings, Decision Theory, and Operational Assumption Violations

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# Historical Context: Statistical Intelligence in WWII

Objective: Infer the unknown total number of tanks produced, $N$, using a small sample of observed serial numbers.

## 1. Espionage Inflation

Conventional military intelligence relied on spy networks and captured prisoners, drastically overestimating production by **4.5–8.2**$\times$ (1,000–1,550 tanks/mo).

## 2. Order Statistics

Allied statisticians analyzed captured chassis and gearbox serials ($k \approx 5$–$50$), mathematically estimating production at **169–327** tanks/mo.

## 3. Speer Ministry Ledgers

Post-war Albert Speer production records confirmed statistical estimates converged within **4%–10%** of actual production.

| Production Month | Conventional Spy Est. | Statistical Est. ($\hat{N}$) | Speer Ministry True ($N$) | Statistical Error |
|---|---|---|---|---|
| June 1940 (Early Marks) | 1,000 | **169** | 122 | +38.5% |
| June 1941 (Panzer III/IV) | 1,550 | **244** | 271 | −**9.9**% |
| August 1942 (Panzer III/IV) | 1,550 | **327** | 342 | −**4.4**% |

**Core Inferential Objective:** *Infer the unknown population ceiling N, moving beyond point estimates to complete uncertainty quantification.*

# Mathematical Formulation: Data, Parameter, and Likelihood

Explicitly separating what is observed from the unknown parameter we seek to infer.

## The Data-Generating Mechanism

Serials drawn by simple random sampling without replacement:

$$S_1, \ldots, S_k \overset{\text{SRSWOR}}{\sim} \{1, 2, \ldots, N\}$$

- **Observed Data:** $D = \{s_1, \ldots, s_k\}$ with sample size $k$.
- **Order Statistics:** $1 \leq s_{(1)} < s_{(2)} < \cdots < s_{(k)} = m$.
- **Sample Maximum:** $m = \max(s_1, \ldots, s_k)$.
- **Physical Boundary:** $N \geq m$ is an absolute physical constraint.

## Inferential Target & Unordered Likelihood

- **Inferential Target:** The target is the unknown population ceiling $N \in \mathbb{N}_{\geq m}$, *not the individual serial numbers*.
- The serial numbers $D$ are observed data; $N$ is the unknown quantity we want to learn about.

$$P(D \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \geq m)$$

- Likelihood is strictly zero for all $N < m$.
- Sample maximum $m$ captures all sample dependence on $N$.

**Key Insight:** *The serial numbers are observed evidence; the population ceiling N is the unknown inferential target.*

# Classical Solution: The UMVUE & Its Limitations

Frequentist point estimation achieves repeated-sampling unbiasedness but provides no posterior distribution.

## The UMVUE Estimator

By the likelihood ratio criterion for minimal sufficiency (Bahadur, 1954), $M = \max_i S_i$ is complete and minimal sufficient.

$$\widehat{N}_{\mathrm{UMVUE}} = m\left(1 + \frac{1}{k}\right) - 1$$

$$\mathrm{Var}(\widehat{N}_{\mathrm{UMVUE}}) = \frac{(N-k)(N+1)}{k(k+2)}$$

- Unbiased across hypothetical repeated samples ($\mathbb{E}[\widehat{N}] = N$).
- Simple closed-form computation requiring only $m$ and $k$.

## Fundamental Limitations of Classical Approach

- **No Posterior Uncertainty Distribution:** The classical approach constructs a point estimator that is unbiased over repeated samples. It does not provide a posterior distribution describing uncertainty about $N$ given this specific sample.
- **Boundary Violations:** Asymptotic Wald intervals ($\widehat{N} \pm 1.96 \cdot \mathrm{SE}$) extend below $m$, placing mass on physically impossible values ($N < m$).
- **Inflexible to Decision Costs:** Cannot adapt to asymmetric operational losses where underestimation carries disastrous strategic costs.

**Transition to Bayes:** *Point estimators provide only an average; Bayesian inference constructs the full posterior distribution $P(N \mid D)$.*

SECTION 01

# Theoretical Properties & Bayesian Foundations

Inferring the posterior distribution, reference priors, loss geometry, and boundary admissibility

# Bayesian Inference: What Are We Trying to Infer?

Establishing the inferential objective: moving from raw identifiers to an epistemic probability distribution.

## 1. What Do We Know?

- The observed sample of serial numbers $D = \{s_1, \ldots, s_k\}$.
- The sample maximum $m = \max(s_1, \ldots, s_k)$.
- Physical fact: the population must satisfy $N \geq m$.

## 2. What Is Unknown?

- The true population ceiling $N \in \{m, m+1, m+2, \ldots\}$.
- We do not know how far above $m$ the true ceiling lies.

## 3. What Does Bayesian Inference Give Us?

A posterior distribution describing the probability of every possible population size, conditional on the observed data:

$$\boxed{P(N = n \mid D)}$$

## 4. What Do We Do With This Distribution?

- Quantify uncertainty via exact credible intervals.
- Derive optimal point estimates tailored to specific loss functions.
- Make principled decisions accounting for risk asymmetry.

**Inferential Pipeline:**

Observed Data $D = \{s_1, \ldots, s_k\}$ $\longrightarrow$ **Posterior Distribution** $P(N = n \mid D)$ $\longrightarrow$ Estimation & Decision Rules $\widehat{N}_L$

**Core Message:** *The posterior distribution itself is the primary inferential output; point estimates are derived afterwards.*

# Ingredients of Bayesian Inference: Likelihood, Prior, Posterior

How observed evidence combines with prior assumptions to quantify uncertainty about $N$.

## 1. Likelihood: Data Evidence

What does the sample tell us about $N$?

$$P(D \mid N = n) = \frac{1}{\binom{n}{k}} \mathbb{I}(n \geq m)$$

- Measures the probability of observing $D$ under each candidate $n$.
- Equals **zero for** $n < m$: automatically assigns zero posterior probability to physically impossible population sizes.

## 2. Prior: Initial Assumptions

What do we assume about $N$ before observing data?

- **Flat Prior ($\pi \propto 1$):** Assumes equal prior weight across all candidate integers.
- **Reciprocal Prior ($\pi \propto 1/n$):** Discrete analog of scale invariance; assigns equal mass to multiplicative decades/octaves ($\sum_{A}^{cA} \frac{1}{n} \approx \log c$).

## 3. Posterior: Updated Belief

What do we believe about $N$ after observing data?

$$P(N = n \mid D) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}$$

- Combines serial evidence with prior structure.
- Quantifies epistemic uncertainty across all physically admissible $n \geq m$.

**Posterior Interpretation:** *The posterior combines sample data with prior assumptions to assign exact probabilities to all candidate population sizes.*

# Discrete Posteriors: Propriety & Moment Existence Thresholds

Different prior assumptions produce different posterior distributions, even when the observed data remain unchanged.

## Exact Discrete Posterior PMFs ($n \geq m$)

- **Flat Prior ($\pi \propto 1$):**

$$P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}$$

Normalizing constant telescopes exactly via negative hypergeometric summation.

- **Reciprocal Prior ($\pi \propto 1/n$):**

$$P(N = n \mid m, k) = \frac{1}{Z(m,k)} \frac{1}{n\binom{n}{k}}$$

$Z(m,k) = \sum_{j=m}^{\infty} \frac{1}{j\binom{j}{k}}$ converges for all $k \geq 1$ (no simple closed form; evaluated numerically).

## Moment Existence Thresholds & Pareto Limit

| Prior Distribution | Proper Posterior | Finite Mean $\mathbb{E}[N]$ | Finite Var $\mathrm{Var}(N)$ |
|---|---|---|---|
| Flat Prior ($\pi \propto 1$) | $k \geq 2$ | $k \geq 3$ ($\mathbb{E} = 2m - 2$) | $k \geq 4$ |
| Reciprocal Prior ($\pi \propto 1/n$) | $k \geq 1$ | $k \geq 2$ ($\mathbb{E} \approx 2m$) | $k \geq 3$ |

**Continuous Pareto Limit ($N \gg k$):**

$$p(N \mid m, k) \approx \frac{(k-1)m^{k-1}}{N^k}, \quad N \geq m$$

- Cumulative CDF: $F(N) = 1 - (m/N)^{k-1}$.
- Median: $\widehat{N}_{\mathrm{med}} \approx m \cdot 2^{1/(k-1)}$.

**Theoretical Takeaway:** *Reciprocal prior guarantees posterior propriety for all $k \geq 1$, whereas flat prior requires $k \geq 2$ for propriety and $k \geq 3$ for finite mean.*

# Estimating $N$ from a Loss Function: Bayesian Decision Theory

A posterior distribution describes uncertainty, but a decision requires us to select a single estimate.

## The Decision-Theoretic Framework

- The optimal estimate depends on the cost assigned to different estimation errors.
- Let $a \in \mathbb{N}_{\geq m}$ denote an estimation action.
- Loss function $L(N, a)$ quantifies the consequence of predicting $a$ when true population is $N$.
- **Posterior Expected Loss (Risk):**

$$R(a) = \mathbb{E}[L(N, a) \mid D] = \sum_{n=m}^{\infty} L(n, a) P(N = n \mid D)$$

- **Optimal Bayes Estimator:**

$$\widehat{N}_L = \arg\min_a R(a)$$

## Example: Squared-Error Loss ($L_2$)

When estimation errors are penalized quadratically:

$$L(N, a) = (N - a)^2$$

$$R(a) = \mathbb{E}[(N - a)^2 \mid D] = \mathrm{Var}(N \mid D) + (\mathbb{E}[N \mid D] - a)^2$$

Setting the derivative to zero yields:

$$\boxed{\widehat{N}_{L_2} = \mathbb{E}[N \mid D] \quad \text{(Posterior Mean)}}$$

- The posterior mean is not "the" Bayesian estimate; it is optimal specifically for squared-error loss.
- If errors carry asymmetric costs, a different estimator is required.

**Core Rule:** *Decisions follow inference: point estimators cannot be chosen without defining a loss function.*

# Loss Functions & Bayesian Estimators: Matching Risk to Mission

Point estimators shift systematically with the decision-maker's loss geometry.

### 1. 0–1 Loss

$L(N, a) = \mathbb{I}(a \neq N)$

$\widehat{N} = \text{Mode} = m$

- Selects most probable single value.
- Severe negative bias ($-16.7\%$ at $k = 5$).

### 2. Absolute ($L_1$)

$L(N, a) = |N - a|$

$\widehat{N} = \text{Median}$

- Linear penalty on deviations.
- Robust against heavy Pareto tails ($-0.9\%$ bias).

### 3. Squared ($L_2$)

$L(N, a) = (N - a)^2$

$\widehat{N} = \mathbb{E}[N \mid D]$

- Quadratic penalty on large errors.
- Pulled rightward by skewed Pareto tail.

### 4. Check Loss $L_q$

$L_q = (N - a)(q - \mathbb{I}(N < a))$

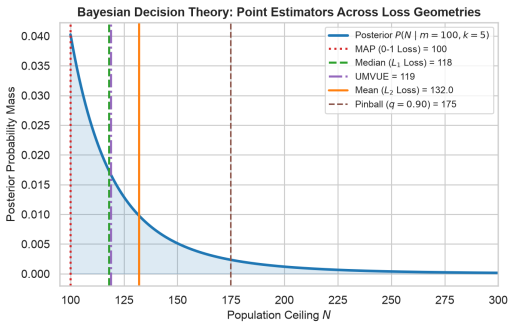
$\widehat{N} = F^{-1}(q \mid D)$

- $q = 0.5$: symmetric median.
- $q = 0.9$: undercounting penalized $9\times$ worse $\Longrightarrow$ defensive safety buffer.

**Military Application:** *In defense intelligence, underestimating enemy armor risks catastrophic defeat; asymmetric check loss ($q = 0.90$) provides a principled operational buffer.*

# Numerical Illustration: Sensitivity of Estimates to Loss

Illustration using the flat-prior posterior, with sample maximum $m = 100$ and sample size $k = 5$.




## Point Estimates Across Objectives ($m = 100, k = 5$)

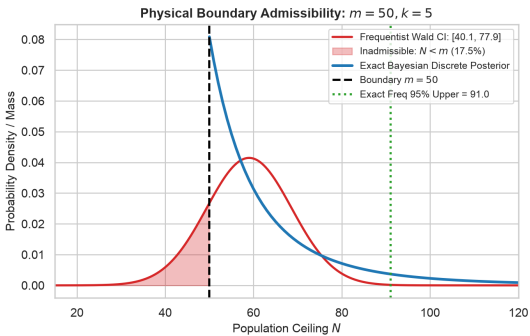
| Objective / Loss | Estimator | Estimate | Strategic Meaning |
|---|---|---|---|
| 0–1 Loss | MAP / Mode | **100** | Extreme lower bound; severe undercount |
| $L_1$ Absolute Loss | Posterior Median | **118** | Minimizes expected absolute error |
| $L_2$ Squared Loss | Posterior Mean | **132** | Quadratic balance; tail-inflated |
| Check Loss ($q = 0.90$) | 90th Percentile | **175** | +75% defensive safety buffer |

- Under the same observed data ($m = 100, k = 5$), optimal estimates span from 100 to 175.
- Posterior expected loss is minimized at a different point for each loss function.

**Key Finding:** *The same observed data can produce vastly different optimal estimates depending on how we value overestimation and underestimation.*

# Physical Boundary Admissibility: Wald vs. Bayesian Intervals

The physical constraint $N \geq m$ must be respected by any valid uncertainty interval.




## Frequentist Wald Boundary Collapse

Wald interval $\widehat{N} \pm 1.96 \cdot \mathrm{SE}$ yields $[40.1, 77.9]$ for $m = 50, k = 5$. It places **17.5% impossible mass** on $N < m = 50$. Asymptotic Gaussianity fails completely near the support boundary.

## Approximate Frequentist Pivotal CI

Large-$N$ pivotal interval $[m, m \cdot \alpha^{-1/k}] = [50.0, 91.0]$ respects the physical boundary by construction.

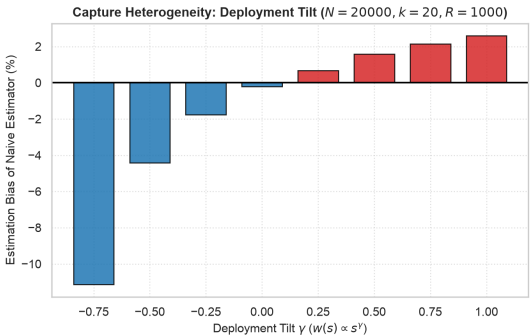
## Bayesian Credible Coherence

Reciprocal-prior 95% credible interval strictly enforces physical admissibility ($P(N < m \mid D) = 0$) and approximately matches the pivotal bound $[50.0, 91.0]$.

**Key Finding:** *The reciprocal-prior posterior respects the physical lower bound and produces credible intervals that approximately agree with the large-N frequentist pivotal interval.*

# Sensitivity to Non-Uniform Sampling: When the Maximum Misleads

Evaluating estimator robustness when capture probability varies with serial age: $w(s) \propto s^{\gamma}$.




## Weighted Capture Mechanics

- $\gamma > 0$ (newer tanks prioritized): pushes $m \to N$, inducing $+1.6\%$ bias at $\gamma = +0.5$.
- $\gamma < 0$ (survivorship / depot attrition): induces $-4.4\%$ bias at $\gamma = -0.5$ and $-11.1\%$ bias at $\gamma = -0.75$.

## Sufficiency Structure Breakdown

- **Known $\gamma$:** $m$ remains sufficient for $N$ via tilted factor $g_N(m; \gamma)$.
- **Unknown $\gamma$:** $m$ alone ceases to be sufficient; joint minimal sufficient statistic is $(m, \sum_{i=1}^{k} \log s_i)$.

**Methodological Warning:** *Applying standard estimators when sampling is tilted introduces systematic bias; joint inference over $(N, \gamma)$ is required.*

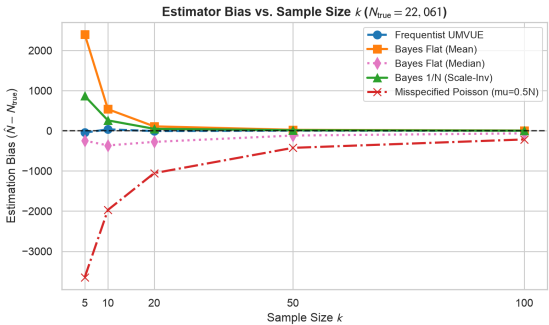
SECTION 02

# Controlled Monte Carlo Simulation Study

Evaluating bias, RMSE contraction, interval calibration, and prior sensitivity across $R = 4,000$ trials ($N = 20,000$)

# Estimator Bias: Finite-Sample Shrinkage vs. Unbiasedness

Evaluating mean bias convergence across $R = 4,000$ Monte Carlo replications on $N_{\text{true}} = 20,000$.




## Frequentist UMVUE Unbiasedness

$\widehat{N} = m(1 + 1/k) - 1$ exhibits near-zero bias across all $k$ ($-48.0 \pm 54.6$ at $k = 5$, $+5.2 \pm 6.0$ at $k = 50$). Unbiased over hypothetical repeated samples.

## Bayes Reciprocal Prior

Posterior Mean has modest bias ($+767.0 \pm 56.8$ at $k = 5$), rapidly decaying to $+13.2$ at $k = 50$. Posterior Median exhibits negative shrinkage ($-901.5$ at $k = 5$).

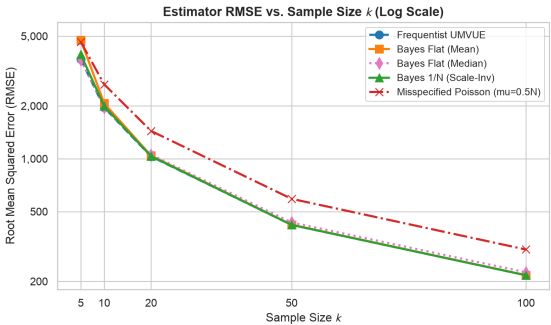
## Flat Prior Tail Inflation

Skewed Pareto tail inflates flat posterior mean by $+2,168.6 \pm 60.7$ at $k = 5$.

**Core Distinction:** *Unbiasedness is a repeated-sampling frequentist property; Bayesian posterior summaries are chosen using the posterior distribution and a decision objective.*

# Faster-Than-Standard-CLT Error Contraction: RMSE & Variance

Boundary order-statistic structure accelerates Root Mean Squared Error decay to $\mathcal{O}(k^{-1})$.




## Theoretical Error Rate

Because $N$ bounds the support, the UMVUE variance satisfies:

$$\mathrm{Var}(\widehat{N}) = \frac{(N-k)(N+1)}{k(k+2)} = \mathcal{O}(k^{-2})$$

giving accelerated rate RMSE $= \mathcal{O}(k^{-1})$, far outperforming standard CLT rate $\mathcal{O}(k^{-1/2})$.

## Empirical Concordance

A $10\times$ increase in $k$ ($5 \to 50$) yields a $\mathbf{9.0}\times$ error reduction ($3,455.2 \to 382.5$), closely matching theoretical $8.6\times$ factor and asymptotic $10\times$ scaling.

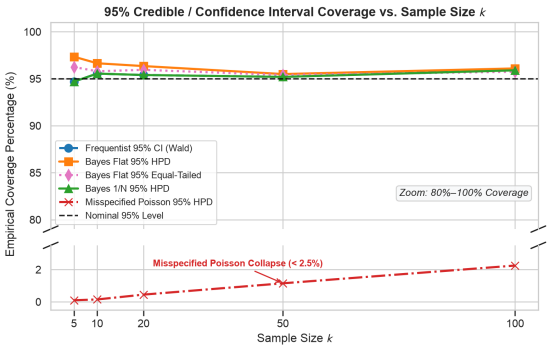
## Estimator Convergence

For $k \geq 50$, all point estimators achieve identical precision ($\approx 383$).

**Contraction Rule:** *The order-statistic boundary structure drives an accelerated $\mathcal{O}(k^{-1})$ RMSE decay, reducing error by nearly an order of magnitude as sample size increases tenfold.*

# Prior Sensitivity: Calibration and the Informative Prior Trap

Evaluating interval reliability and the catastrophic consequences of dogmatic prior misspecification.




## Reciprocal Prior Calibration

Empirical coverage closely tracks nominal 95%: **94.4%** $\pm$ **0.36%** at $k = 5$ and **95.6%** $\pm$ **0.32%** at $k = 50$.

## Informative Boundary Trapping

A dogmatically narrow misspecified Poisson($\mu = 10k$) prior traps posterior against the boundary $[m, m + 5]$, collapsing coverage to **0.3%**.

## Hierarchical Hyperprior Resolution

Gamma hyperprior over Poisson rate ($\mathrm{CV} \geq 1$) provides adaptive dispersion, restoring coverage to **94.6%–95.0%** even under 50% prior mean error.

**Calibration Finding:** *The reciprocal prior produces empirical coverage close to 95% in our simulation. A misspecified Poisson prior causes severe collapse, while a hierarchical prior restores calibration.*

# Monte Carlo Benchmark: Empirical Performance Leaderboard

Multi-dimensional evaluation across $R = 4,000$ replications on $N_{\text{true}} = 20,000$ (Seed 42).

| Framework | Estimator | $k$ | Mean Bias $\pm$ SE | RMSE (Emp / Th) | 95% Cov $\pm$ SE | Mean Width |
|---|---|---|---|---|---|---|
| UMVUE (Frequentist) | $\widehat{N} = m(1 + 1/k) - 1$ | 5 | $-48.0 \pm 54.6$ | 3,455.2 / 3,380.3 | 90.0% $\pm$ 0.47% (Wald) | 13,219 |
| Approximate Pivotal CI | Large-$N$ Pivotal Bound | 5 | — | — | **94.4%** $\pm$ 0.36% | 13,644 |
| Bayes Flat Prior | Posterior Mean | 5 | $+2,168.6 \pm 60.7$ | 4,409.0 / 4,363.9 | **97.4%** $\pm$ 0.25% | 18,535 |
| Bayes Flat Prior | Posterior Median | 5 | $-226.5 \pm 54.1$ | 3,431.3 / — | **97.4%** $\pm$ 0.25% | 18,535 |
| Bayes Reciprocal $(1/n)$ | Posterior Mean | 5 | $+767.0 \pm 56.8$ | 3,671.1 / 3,618.6 | **94.4%** $\pm$ 0.36% | 13,622 |
| Bayes Reciprocal $(1/n)$ | Posterior Median | 5 | $-901.5 \pm 52.3$ | **3,427.5** / — | **94.4%** $\pm$ 0.36% | 13,622 |
| UMVUE (Frequentist) | $\widehat{N} = m(1 + 1/k) - 1$ | 50 | $+5.2 \pm 6.0$ | 382.5 / 391.8 | 95.1% $\pm$ 0.34% (Wald) | 1,536 |
| Approximate Pivotal CI | Large-$N$ Pivotal Bound | 50 | — | — | **95.6%** $\pm$ 0.32% | 1,211 |
| Bayes Reciprocal $(1/n)$ | Posterior Mean | 50 | $+13.2 \pm 6.0$ | 382.8 / 392.0 | **95.6%** $\pm$ 0.32% | 1,209 |
| Bayes Flat Prior | Posterior Mean | 50 | $+21.6 \pm 6.1$ | 383.4 / 392.4 | **95.9%** $\pm$ 0.31% | 1,237 |

## 1. Point Accuracy

UMVUE is exactly mean-unbiased; Bayes reciprocal median achieves lowest small-sample RMSE $(3,427.5)$.

## 2. Interval Reliability

Reciprocal credible interval achieves 94.4% coverage; Wald interval suffers from boundary distortion (90.0%).

## 3. Interval Precision

Reciprocal intervals are substantially sharper $(13,622)$ than flat prior intervals $(18,535)$.

**Synthesis:** *The reciprocal prior provides a useful balance between estimation accuracy, empirical calibration, and physical admissibility in the simulated setting.*

SECTION 03

# Real-World Empirical Stress-Testing

Evaluating textbook assumptions against WWII archives and 541,909 commercial transactions

# Empirical Stress-Testing: Two Real-World Datasets

Examining what happens when textbook assumptions meet physical operational systems.

## 1. Archival WWII German Tank Ledgers

- **Data Source:** Waffenamt ordnance ledgers and Speer Ministry production books.
- **Identifiers Analyzed:** Chassis serial blocks and sequential ZF gearbox stampings.
- **Operational Reality:** Pre-allocated factory blocks; chassis diverted to assault guns (StuG III/IV).

## 2. Commercial Invoice Ledger (Online Retail)

- **Data Source:** 541,909 transaction line items across 4,372 customer accounts.
- **Serials Analyzed: 22,061 distinct sequential sales invoices**.
- **Sequence Address Span:** $[536365, 581587] \implies \mathbf{45,223}$ slots.
- **Integer Voids:** 23,162 unrecorded slots (**51.2% void rate**).

**Crucial Methodological Clarification:** These datasets are used to examine the **limitations of sequential-number inference in operational settings**, NOT to claim that both datasets satisfy the German Tank sampling model.

**Distinction:** *541,909 represents raw transaction line items; 22,061 represents distinct sales invoice serial numbers.*

# Finding 1: Chronological Generation vs. Uniform Sampling

Real-world serials are generated dynamically over time, not drawn as static random samples.

## What Was Measured & Calculated

- **Measured:** Arrival timestamps and sequential numeric identifiers across all recorded transactions.
- **Calculated:** Weekly arrival volume $\Delta s / \Delta t$ and cumulative serial growth trajectories.
- **Observed Stream Dynamics:** Weekly volume accelerates from $\approx$ **370** invoices/week in Q1 to $>$ **790** invoices/week in Q4, peaking at **894** invoices/week in mid-November.

## Research Question Impact & Statistical Conclusions

- **Textbook Assumption:** Assumes a *static population* from which observations are sampled randomly.
- **Operational Reality:** Invoices form an open, chronological transaction stream with non-stationary volume.
- **Exchangeability Violation:** Chronological ordering induces strong temporal autocorrelation. Successive observations are not exchangeable in the way the textbook model assumes.

**Statistical Conclusion:** *A static population-ceiling model cannot automatically be applied to a dynamic transaction stream.*

# Finding 2: Integer Voids & System Architecture

Operational workflows generate massive integer voids (51.2% unrecorded rate).

## Measured Sequence Breakdown

- **Total Integer Address Slots: 45, 223** ($s_{\max} - s_{\min} + 1 = 581,587 - 536,365 + 1$).
- **Recorded Sales Invoices: 22, 061** (48.78% recorded).
- **Unrecorded Slots (Voids): 23, 162** (**51.22**% void rate).
- **Gap Distribution:** Median gap $\Delta s = 1.0$, mean 2.05, max 76.

## Verified Facts vs. Operational Hypotheses

- **Verified Cancellations (3,836):** Exactly 3,836 C-prefixed invoice numbers exist in the raw dataset, directly explaining **16.56% of voids**.
- **Operational Hypotheses (19,326):** Without internal server logs, the remaining 83.44% voids plausibly reflect uncommitted checkout sessions, gateway dropouts, or distributed sequence caching.
- **Wartime Parallel:** Waffenamt pre-allocated chassis blocks; assembly diversions created sequence gaps.

**Crucial Distinction:** *The 3,836 cancellations are verified in data; the remaining 19,326 voids are plausible operational hypotheses, not verified findings.*

# Finding 3: Goodness-of-Fit & The Uniformity Fallacy

Uniform dispersion across an address span is not exchangeability.

## What Was Measured & Calculated

- Normalized span positions $u_i = (s_i - s_{\min})/(s_{\max} - s_{\min})$ evaluated against uniform CDF $F(u) = u$.
- **Maximum Absolute Discrepancy:** $D = \sup_u |F_n(u) - u| = 0.012$ $(n = 22,061)$.
- **Invalidity of Standard KS Table:** Estimating endpoints from data pins $u_{(1)} = 0$ and $u_{(n)} = 1$. Textbook KS distributions do not apply; formal testing requires parametric bootstrap.

## What the ECDF Tells Us vs. What It Does Not

| What the ECDF Tells Us | What It Does NOT Tell Us |
|---|---|
| Even distribution across span | Whether observations are exchangeable |
| Absence of massive block gaps | Whether sampling mechanism is uniform |
| Descriptive numerical regularity | Validity of the German Tank model |

- A linear ECDF merely shows even allocation across the year.
- Successive observations are strongly autocorrelated in time.

**Methodological Rule:** *Never interpret a non-significant goodness-of-fit test as proof of exchangeable uniform sampling.*

# Finding 4: Estimand Disambiguation & Thinning Models

What exactly are we trying to count? Sequence capacity vs. realized transaction volume.

## Estimand Disambiguation

- **Estimand A ($N_{\text{span}} = 45,223$):** Total allocated address capacity ($s_{\max} - s_{\min} + 1$). Raw estimator without base offset yields $\approx 593k$.
- **Estimand B ($N_{\text{count}} = 22,061$):** Net volume of realized, active completed items. Shifting by $s_{\min}$ gives $+105\%$ error over active count.
- **Oracle Trap:** Rank re-indexing requires knowing all valid keys in advance (un-deployable in live audits).

## The Bernoulli Thinning Model & Caveats

$$N_{\text{count}} \mid N_{\text{span}}, \rho \sim \text{Binomial}(N_{\text{span}}, \rho)$$

- **Retrospective Calculation:** Using the complete dataset ($k = 22,061$), $\widehat{\rho}_{\text{retrospective}} = \frac{22,060}{45,222} \approx 0.4878$.
- **Operational Identifiability Caveat:** This calculation uses the complete observed invoice set. In an incomplete field sample ($k \ll N$), the sample spacing ratio estimates sample density, not $\rho$.
- Estimating both $N_{\text{span}}$ and $\rho$ from an incomplete sample requires auxiliary data.

**Conceptual Takeaway:** *Bernoulli thinning clarifies the estimand distinction conceptually; estimating both capacity and void rate requires auxiliary identifying structure.*

# Connections to Other Areas of Statistical Inference

How discrete order-statistic inference connects across domains, and where its boundaries lie.

## Conceptual Cross-Domain Connections

- **Dynamic Production:** Monotonic state-space models ($N_t \geq N_{t-1}$) enforce output non-negativity across periods.
- **Software Reliability:** Jelinski–Moranda defect exhaustion models mirror serial discovery for software release readiness.
- **Aviation Safety:** Beta–Bernoulli mixtures detect cloned counterfeit serial numbers in critical parts supply chains.
- **Wildlife Abundance:** Two-stage cascades infer tag loss before estimating population size via mark-recapture.

## Methodological Boundaries

- **Non-Sequential Keys:** Randomized UUIDv4 or cryptographic hashes completely destroy order-statistic sufficiency.
- **Non-Random Gaps:** Classified blocks or unrecorded overseas entities require informative gap priors.
- **Dynamic Streams:** Fast-changing arrival rates require time-varying Poisson arrival processes.
- **Identifiability Limits:** Incomplete samples cannot simultaneously identify capacity and void rate without auxiliary data.

**Synthesis:** *Order-statistic inference generalizes broadly across domains, provided analysts explicitly model data-generating mechanisms and respect identifiability boundaries.*

SUMMARY & CONCLUSIONS

# Key Takeaways for Bayesian Discrete Inference

## 1. Objective Foundations

The target is the discrete ceiling $N$, with evidence captured by sample maximum $m$. Scale-invariant reciprocal prior $\pi(N) \propto 1/n$ yields proper posteriors ($k \geq 1$), respects physical boundaries ($N \geq m$), and matches the large-$N$ pivotal bound.

## 2. Decision Theory & Rates

Point estimation follows loss geometry: median minimizes absolute error; check loss ($q = 0.9$) builds defensive buffers. Extreme-value order statistics produce accelerated $\mathcal{O}(k^{-1})$ RMSE contraction ($9\times$ reduction for $10\times$ sample).

## 3. Real-World Reality

Operational data violate textbook assumptions via chronological streaming and integer voids (51.2%). Bernoulli thinning disambiguates address capacity from realized transaction volume.

**Central Research Answer:** Bayesian inference provides a principled framework for quantifying uncertainty about an unknown discrete population ceiling, selecting estimates under mission-specific decision objectives, and evaluating inferential reliability. Its practical validity in the physical world, however, depends entirely on whether operational data-generation mechanisms satisfy model assumptions.

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